Bonus Guide 2 of 5

THE POTTERY TEMPLATE VAULT

Glazes Guide

Types, application methods & troubleshooting for ceramic glazes

Glazes Guide

> A glaze is a glass-like coating fused to the clay body during firing. It seals the piece, makes it food-safe, and gives it color, texture and personality. This guide covers what you need to know to choose and apply glazes confidently.

Contents

1. What glaze is — the basics
2. Types of glaze
3. Reading a glaze label
4. Preparing commercial glazes
5. Application methods
6. Glaze thickness and why it matters
7. Wax resist and masking
8. Glaze and food safety
9. Troubleshooting common glaze problems

1 · What Glaze Is

Glaze is made of three things: **silica** (glass-former), **alumina** (stabilizer) and **flux** (melting agent). When fired, they melt together and fuse to the clay surface, forming a glass coating.

The final result depends on:

- The glaze recipe (ingredients and proportions)
- The firing temperature and atmosphere (oxidation or reduction)
- The thickness of the application
- The clay body underneath

Because all four variables interact, two potters can use the same glaze and get different results. Test tiles are essential.

2 · Types of Glaze

Type	Appearance	Best for
Glossy	Shiny, smooth, light-reflective	Tableware, mugs, bowls — easy to clean
Matte / satin	Low sheen, velvety or soft surface	Decorative pieces, vases, modern aesthetic
Celadon	Translucent green-grey, pools in texture	Carved work — glaze collects in carved areas
Tenmoku	Dark brown to black, breaks to rust at edges	Teapots, bowls, simple forms
Shino	White-orange, dry, carbon-trap effects	Rustic, organic pieces
Crystalline	Macro-crystals grow on the surface during firing	Advanced technique — requires precise firing schedule
Ash glaze	Variable, earthy, streaky	Wood-fire or salt-fire aesthetics
Underglaze	Matte or slight sheen, applied under a clear topcoat	Painted decoration, sgraffito

3 · Reading a Glaze Label

- **Temperature range** — listed as a cone number (e.g. Cone 6, Cone 10) or temperature (e.g. 1220–1260 °C). Must match your kiln and clay body.
- **Firing atmosphere** — "oxidation" (electric kiln) or "reduction" (gas kiln). Some glazes only develop their full color in reduction.
- **Food-safe** — look for this explicit statement on commercial glazes. If it is not stated, assume it is not.
- **Application method** — brushing, dipping, or pouring. Many commercial brushing glazes are not thin enough for dipping without dilution.

4 · Preparing Commercial Glazes

Brushing glazes (from a jar)

- Stir thoroughly before every use — the pigments and materials settle.
- If the glaze has thickened in storage, add a few drops of water and stir well.
- Apply 2–3 coats, letting each dry before the next.

Dry/powdered glazes (mixed in the studio)

- Always wear a dust mask when handling dry glaze materials.
- Mix glaze by weight according to the recipe. Add water gradually to reach the consistency of milk (for dipping) or single cream (for brushing).

- Sieve through an 80-mesh screen to remove lumps.
- Let the glaze rest 24 hours before use; sieve again before applying.

> ⚠️ **Warning**
> Never eat, drink or touch your face while handling raw glaze materials. Wash hands thoroughly after glazing. Many glaze materials are toxic in raw form — they are safe only after firing.

5 · Application Methods

Brushing

The most accessible method. Use a wide, soft brush (fan brush or flat wash brush). Apply in smooth strokes, always in the same direction per coat. Alternate direction 90° on the second coat for even coverage.

- 3 coats on vertical surfaces
- 2 coats on horizontal surfaces (glaze drips less)
- Let each coat go chalky-dry before applying the next

Dipping

Dip the bisqueware piece into a bucket of glaze. Hold it by the foot and dip for 3–5 seconds. Wipe the foot clean immediately. Very fast and even, but requires a large enough bucket to submerge the piece.

Pouring

Pour glaze over the outside of the piece while rotating it. Fill the inside, swirl, then pour out the excess. Good for large pieces. The interior and exterior can be glazed with different glazes.

Spraying

Requires a spray gun, compressor and spray booth with ventilation. Gives very even, thin coats. Ideal for gradients and layering multiple colors.

6 · Glaze Thickness and Why It Matters

Glaze thickness is measured with a pin tool — insert it into the wet glaze, pull it out, and measure how much glaze coats the pin. A standard thickness is **1.5–2 mm** for most cone 6 glazes.

Too thin	Correct	Too thick
Dry, powdery surface after firing; weak color; may flake	Smooth, even, full color development	Runs down the piece; sticks to the kiln shelf; blistering

> 💡 **Tip**
>
> Test your glaze on a vertical test tile and fire it alongside your work. If the glaze runs, it is too thick or too fluid — thin it with water or reduce application to 2 coats.

7 · Wax Resist and Masking

Apply wax emulsion (hot wax or water-based wax resist) with a brush to areas you do not want glazed. The wax repels the glaze. After firing, the wax burns away, leaving the clay body exposed.

Uses

- **Foot rings** — always wax the foot before glazing so the piece does not stick to the kiln shelf.
- **Design elements** — wax a pattern, glaze over it; glaze sticks everywhere except the waxed area.
- **Two-glaze separation** — wax the lower third of a piece, dip in glaze 1 (upper), let dry, remove wax, dip in glaze 2 (lower). Clean separation.

> ⚠️ **Warning**
>
> Hot wax requires a wax pot kept at 65–70 °C. It is a fire hazard if overheated. Water-based wax resist is safer and works just as well for most purposes.

8 · Glaze and Food Safety

A glaze is food-safe when it is:

- Fully fired to the correct temperature (under-fired glazes can leach metals)
- Free of lead, barium and cadmium (check the manufacturer's safety data sheet)
- Applied with enough thickness to form a complete glass layer
- Free of crawling, crazing or pinholes on food-contact surfaces

> ⭐ **Important**
>
> Commercial glazes labeled "food safe" are reliable when fired correctly. If in doubt, test with an FDA leach test kit or have the piece tested by a lab before use as tableware.

9 · Troubleshooting Common Glaze Problems

Problem	Cause	Fix
Crawling (glaze pulls away in patches)	Glaze applied too thick, or on dusty bisque	Apply thinner; wipe bisque clean before glazing
Crazing (fine cracks in the glaze surface)	Glaze shrinks more than the clay body	Adjust the glaze recipe or switch to a glaze matched to your clay body
Pinholing (tiny holes in the surface)	Gas escaping during firing	Slow the firing through 900–1100 °C; fire to the correct top temperature
Running / drips	Glaze too thick or too fluid; over-fired	Thin the glaze; apply fewer coats; check firing schedule
Dull, dry surface	Under-fired; glaze too thin	Refire or apply a thicker coat next time
Color different from test tile	Different firing atmosphere or temperature	Check kiln calibration; compare atmosphere (oxidation vs reduction)

Glazing checklist

- Is the bisqueware clean and dust-free?
- Is the glaze stirred and at the right consistency?
- Have I waxed the foot ring?
- Did I apply 2–3 even coats, each fully dry before the next?
- Is the foot completely clean of glaze so it will not stick to the kiln shelf?

Next: Engobes, Pigments & Oxides Guide — adding color to clay before and after bisque firing.